Shadow Mirror: A Consent-Based Probabilistic Framework for Computational Self-Reflection and Human Behavioral Modeling

Abstract

Understanding human cognition requires more than predicting observable behavior. Shadow Mirror is a consent-based probabilistic framework that models relationships among self-report, demonstrated performance, observable behavior, emotional state, and contextual evidence. Rather than assigning deterministic labels, it generates uncertainty-aware hypotheses that are transformed into reflective questions. Ethics are embedded through informed consent, uncertainty estimation, temporal decay, trauma-aware safeguards, local data ownership, and interpretable Bayesian reasoning.

Introduction

Current AI profiling systems often rely on static labels and opaque inference. Shadow Mirror instead studies contradictions between beliefs, abilities, and actions while explicitly acknowledging uncertainty and change.

Conceptual Foundations

The framework integrates three evidence streams: self-report, performance, and behavior. Contradictions between these streams become signals for structured reflection instead of errors.

Research Objectives

The system investigates what people think they know, what they demonstrate, what they repeatedly do, and how these relationships evolve over time.

Bayesian Reflection Engine

A Bayesian inference engine maintains posterior probabilities over competing hypotheses (e.g., calibrated, underconfident, overconfident). Evidence updates these probabilities continuously.

Temporal Modeling

Observations are timestamped and gradually decay in influence, ensuring outdated evidence does not dominate the model.

Reflective Outputs

Instead of identity labels, Shadow Mirror produces reflective questions that encourage self-examination.

Ethical Architecture

Two modes exist: Self Mirror and Consented Study. Consent is explicit, revocable, and enforced by software.

Safety Mechanisms

Trauma-aware reasoning, evidence thresholds, context preservation, and uncertainty reporting reduce overinterpretation.

Data Governance

Local storage, exportability, deletion, and user ownership preserve privacy.

Software Architecture

Modules include inference, pattern analysis, reflection generation, consent management, safety, persistence, and a command-line interface.

Validation

Automated tests verify Bayesian updating, consent enforcement, uncertainty estimation, safety rules, temporal reasoning, and persistence.

Applications

Potential applications include mental health support, education, coaching, organizational psychology, and human-centered AI.

Limitations

The framework does not diagnose disorders, replace clinicians, or perform covert profiling.

Future Research

Future directions include physiological sensing, hierarchical Bayesian models, causal inference, longitudinal validation, and LLM-assisted reflection.

Conclusion

Shadow Mirror proposes a new paradigm for reflective AI that prioritizes probabilistic reasoning, ethical safeguards, and human-centered self-reflection over deterministic profiling.